

# Nitrogen (N<sub>2</sub>) Purge Simulation for Hydrogen (H<sub>2</sub>) Line Maintenance

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## 1. Chemical Engineering & Safety Context

Before performing hot work (welding, cutting, grinding, or thermal line modifications) on a hydrogen pipeline, the residual combustible gas must be displaced by an inert gas—typically

nitrogen (N<sub>2</sub>)—until the concentration is reduced well below the **Lower Explosive Limit (LEL)**.

## Hydrogen Flammability Envelope in Air

| Parameter / Flammability Threshold                | Value (% vol)       | Concentration (ppm) | Chemical Engineering Context                                       |
|---------------------------------------------------|---------------------|---------------------|--------------------------------------------------------------------|
| <b>Lower Explosive Limit (LEL)</b>                | 4.0%                | 40,000 ppm          | Minimum concentration in air capable of sustaining combustion      |
| <b>Upper Explosive Limit (UEL)</b>                | 75.0%               | 750,000 ppm         | Maximum concentration in air for flame propagation                 |
| <b>Limiting Oxygen Concentration (LOC)</b>        | 5.0% O <sub>2</sub> | 50,000 ppm          | Minimum O <sub>2</sub> required to support H <sub>2</sub> ignition |
| <b>Safe Limit for Hot Work</b>                    | ≤ 0.5%              | 5,000 ppm           | Industry standard safety threshold (1/8 of LEL or < 10% of LEL)    |
| <b>Initial Line Concentration (C<sub>0</sub>)</b> | 100.0%              | 1,000,000 ppm       | Undiluted hydrogen gas prior to maintenance purge                  |

- NFPA 86: Standard for Ovens and Furnaces (purge requirements)
- API 2009: Safe Welding, Cutting, and Hot Work Practices
- Perry's Chemical Engineers' Handbook, 9th Ed.: Section 27 — Safety
- PYroMat Handbook: Chapter 8 — Multi-Phase Model (mp1) for N<sub>2</sub> properties

## Fundamental Purging Dynamics

In an ideal, unmixed displacement process (plug flow), purging requires exactly 1 line volume of inert gas. However, in continuous displacement with mixing, concentration decay follows an exponential relationship:

$$C = C_0 \cdot e^{-N}$$

where  $N$  is the number of purge line volumes exchanged. To dilute pure H<sub>2</sub> (100%) down to a safe working threshold (0.4% – 0.5%), the theoretical minimum purge volume is:

$$N = \ln\left(\frac{100.0}{0.4}\right) = \ln(250) \approx 5.52 \text{ line volumes}$$

As established in industrial safety guidelines (ref: [cv.ukhan.org/posts/h2\\_purge/](https://cv.ukhan.org/posts/h2_purge/)), standard field procedures mandate **7 to 10 line volume exchanges** to account for non-ideal mixing ( $\eta < 1.0$ ), stagnant pockets, and dead-leg fittings.

## Physical Hazards Modeled in this Notebook

This simulation evaluates two critical physical phenomena:

1. **Joule-Thomson (JT) Isenthalpic Cooling:** Rapid expansion of N<sub>2</sub> from high-pressure cylinders (200 bar) to line pressure (1.5 bar) causes severe temperature drops ( $\Delta T \approx -30^\circ\text{C}$ )

- ). The resulting cold gas can cool carbon steel piping near or below its Minimum Design Metal Temperature (MDMT), inducing brittle fracture risks.
2. **Mixing Efficiency ( $\eta$ ) Effects:** Real purge operations deviate from ideal plug flow and perfect CSTR mixing. We model dilution kinetics via  $C = C_0(1 - \eta)^N$  across realistic mixing efficiencies ( $\eta = 0.3 - 1.0$ ).

## Environment Setup & Thermodynamic Unit Configuration

We begin by loading scientific computing libraries ( `numpy` , `matplotlib` ) and the **PYroMat** thermodynamic package. PYroMat unit settings are explicitly set to metric process engineering standards: `bar` for pressure, `K` for temperature, and `kJ` for specific enthalpy/energy.

```
In [1]: import pyromat as pm
import numpy as np
import matplotlib.pyplot as plt
import warnings

warnings.filterwarnings('ignore')

# Configure PYroMat thermodynamic units
pm.config['unit_pressure'] = 'bar'
pm.config['unit_temperature'] = 'K'
pm.config['unit_energy'] = 'kJ'

print("PYroMat environment initialized successfully:")
print(f"  Pressure unit:    {pm.config['unit_pressure']}")
print(f"  Temperature unit: {pm.config['unit_temperature']}")
print(f"  Energy unit:      {pm.config['unit_energy']}\n")

%reload_ext watermark
%watermark -v -m -p PYroMat,numpy,matplotlib
```

PYroMat environment initialized successfully:

Pressure unit: bar  
Temperature unit: K  
Energy unit: kJ

Python implementation: CPython

Python version : 3.11.2

IPython version : 8.5.0

PYroMat : 2.2.6

numpy : 1.26.4

matplotlib: 3.10.9

Compiler : GCC 12.2.0

OS : Linux

Release : 6.12.93+rpt-rpi-v8

Machine : aarch64

Processor :

CPU cores : 4

Architecture: 64bit

```
/home/ukhan/.local/lib/python3.11/site-packages/matplotlib/projections/__init__.py:63: UserWarning: Unable to import Axes3D. This may be due to multiple versions of Matplotlib being installed (e.g. as a system package and as a pip package). As a result, the 3D projection is not available.
  warnings.warn("Unable to import Axes3D. This may be due to multiple versions of "
```

## 2. Thermodynamic Gas Models

To accurately capture fluid behavior, we employ distinct thermodynamic models for the inert purge gas ( $\text{N}_2$ ) and the displaced gas ( $\text{H}_2$ ).

### Real Gas Equation of State for Nitrogen ( `mp.N2` )

Nitrogen at cylinder storage conditions (200 bar) exhibits significant real-gas deviations from the ideal gas law ( $Z = \frac{pv}{RT} \neq 1$ ). We utilize PYroMat's **mp1 (multi-phase)** model, which implements the empirical Helmholtz free energy equation of state (Span–Wagner formulation):

$$\phi(\delta, \tau) = \phi^o(\delta, \tau) + \phi^r(\delta, \tau)$$

where  $\delta = \rho/\rho_c$  is the reduced density and  $\tau = T_c/T$  is the inverse reduced temperature. This real-gas formulation is crucial for:

- Calculating exact real-gas density  $\rho_{\text{N}_2}(T, p)$  across wide pressure drops.
- Evaluating the Joule-Thomson coefficient  $\mu_{\text{JT}} = \left(\frac{\partial T}{\partial p}\right)_h$ , which is strictly **zero** for an ideal gas but positive ( $\approx 0.2 \text{ K/bar}$ ) for real  $\text{N}_2$  near ambient temperatures.

### Ideal Gas Model for Hydrogen ( `ig.H2` )

Hydrogen is modeled using PYroMat's `ig.H2` class. Because the pipeline operates at low pressure (1.5 bar(a)) and hydrogen exhibits near-unity compressibility at ambient temperatures, the ideal gas EOS provides sufficient accuracy for mass balance and concentration decay calculations.

```
In [2]: # Load substance models
n2 = pm.get('mp.N2')      # Real gas model – Helmholtz EOS for accurate JT & density
h2 = pm.get('ig.H2')      # Ideal gas model – adequate for low-pressure H2

print(f"Nitrogen substance model: {n2.sid()} (Class: {n2.data['class']})")
print(f"Hydrogen substance model: {h2.sid()} (Class: {h2.data['class']})")
```

```
Nitrogen substance model: mp.N2 (Class: mp1)
Hydrogen substance model: ig.H2 (Class: ig2)
```

## 3. Pipeline Specifications & Purging Flow Dynamics

### Mechanical & Process Parameters

We model a standard industrial hydrogen transfer line in a chlor-alkali / petrochemical facility:

- **Nominal Diameter:** 6-inch Schedule 40 carbon steel pipe ( $D_{\text{inner}} = 0.15 \text{ m}$ ).

- **Length:**  $L = 50.0$  m.
- **Pipeline Volume:**  $V_{\text{line}} = \frac{\pi}{4} D_{\text{inner}}^2 L = \frac{\pi}{4} (0.15)^2 (50) \approx 0.8836 \text{ m}^3$  (883.6 L).
- **Initial State:** Pure hydrogen ( $C_0 = 100.0\%$ ), line pressure  $P_{\text{line}} = 1.5$  bar(a), ambient temperature  $T_{\text{line}} = 303.15$  K (30°C).

## Purge Supply & Volumetric Flow Conversion

Purge gas is supplied at  $50.0 \text{ Nm}^3/\text{h}$  from a 200 bar cylinder manifold regulated down to 3.0 bar(a).

The normal flow rate  $\dot{V}_{\text{norm}}$  at  $P_{\text{norm}} = 1.01325$  bar,  $T_{\text{norm}} = 273.15$  K) is converted to actual volumetric flow rate  $\dot{V}_{\text{actual}}$  at pipeline conditions using the ideal gas relationship:

$$\dot{V}_{\text{actual}} = \dot{V}_{\text{norm}} \cdot \left( \frac{P_{\text{norm}}}{P_{\text{line}}} \right) \cdot \left( \frac{T_{\text{line}}}{T_{\text{norm}}} \right)$$

The time required to exchange one complete pipeline volume is:

$$t_{\text{vol}} = \frac{V_{\text{line}}}{\dot{V}_{\text{actual}}}$$

```
In [3]: # Hydrogen line specifications
LINE_LENGTH = 50.0          # m
LINE_DIAMETER = 0.15        # m (6 inch Schedule 40)
LINE_VOLUME = np.pi * (LINE_DIAMETER/2)**2 * LINE_LENGTH # m³
LINE_VOLUME_L = LINE_VOLUME * 1000 # Litres
LINE_PRESSURE = 1.5         # bar(a) – typical H₂ supply pressure
LINE_TEMP = 303.15         # K (30 °C)

# Initial H₂ concentration & target safety limits
INITIAL_H2_PCT = 100.0      # % vol (pure H₂ initially)
TARGET_H2_PCT = 0.5        # % vol (target limit for welding/hot work)
LEL_H2 = 4.0               # % vol (Lower Explosive Limit in air)
SAFE_MARGIN = 0.125        # 1/8 of LEL threshold

# Purge gas (N₂) supply parameters
N2_CYLINDER_PRESSURE = 200.0 # bar(a)
N2_REGULATOR_SET = 3.0      # bar(a)
N2_TEMPERATURE = 303.15     # K (30 °C)

# Purge flow rate parameters
PURGE_FLOW_NM3H = 50.0      # Nm³/h
P_NORM = 1.01325           # bar
T_NORM = 273.15            # K

# Convert Nm³/h to actual m³/s at line operating conditions
PURGE_FLOW_ACTUAL = (PURGE_FLOW_NM3H / 3600.0) * (P_NORM / LINE_PRESSURE) * (LINE_TEMP /
TIME_PER_VOLUME = LINE_VOLUME / PURGE_FLOW_ACTUAL # seconds per exchange

# Mixing efficiencies to evaluate
MIXING_EFFICIENCIES = [0.3, 0.5, 0.7, 1.0]

print(f"Pipeline volume:          {LINE_VOLUME_L:.1f} L ({LINE_VOLUME:.4f} m³")
```



```
print(f"Actual flow rate: {PURGE_FLOW_ACTUAL:.4f} m³/s")
print(f"Time per volume exchange: {TIME_PER_VOLUME:.1f} s ({TIME_PER_VOLUME/60.0:.2f} mi
```

```
Pipeline volume:      883.6 L (0.8836 m³)
Actual flow rate:     0.0104 m³/s
Time per volume exchange: 84.9 s (1.41 min)
```

## 4. Joule-Thomson Cooling Analysis & Numerical Perturbation

### Thermodynamics of Isenthalpic Let-Down

When compressed nitrogen undergoes throttling across a pressure reduction valve (isenthalpic expansion,  $\Delta h = 0$ ), its temperature change is governed by the Joule-Thomson coefficient  $\mu_{JT}$ :

$$\mu_{JT} \equiv \left( \frac{\partial T}{\partial p} \right)_h = -\frac{1}{c_p} \left( \frac{\partial h}{\partial p} \right)_T = \frac{1}{c_p} \left[ T \left( \frac{\partial v}{\partial T} \right)_p - v \right]$$

For an ideal gas,  $T \left( \frac{\partial v}{\partial T} \right)_p = v$ , rendering  $\mu_{JT} = 0$ . For real nitrogen at ambient temperature, attractive intermolecular forces dominate during expansion, absorbing kinetic energy and producing a substantial cooling effect ( $\mu_{JT} > 0$ ).

### Numerical Perturbation Method for Property Derivatives

As discussed in [PYroMat Issue #71](#), PYroMat's thermodynamic state objects do not provide explicit analytical derivatives of enthalpy with respect to pressure. We compute  $\left( \frac{\partial h}{\partial p} \right)_T$  using a central-difference numerical perturbation:

$$\left( \frac{\partial h}{\partial p} \right)_T \approx \frac{h(T, p + \varepsilon p) - h(T, p - \varepsilon p)}{2\varepsilon p}$$

where  $\varepsilon = 10^{-6}$  is a non-dimensional perturbation fraction. The Joule-Thomson coefficient is then evaluated as:

$$\mu_{JT}(T, p) = -\frac{h(T, p + \varepsilon p) - h(T, p - \varepsilon p)}{2\varepsilon p \cdot c_p(T, p)}$$

### Two-Stage Let-Down Thermal Profile

The pressure let-down occurs in two distinct stages:

1. **Stage 1 (Regulator):** 200.0 bar  $\rightarrow$  3.0 bar
2. **Stage 2 (Line Let-Down):** 3.0 bar  $\rightarrow$  1.5 bar

The final downstream temperature  $T_2$  is solved directly by inverting the real-gas enthalpy relation using `mp.N2`:

$$h(T_2, p_2) = h(T_1, p_1)$$

```

In [4]: def jt_coeff(substance, T, p):
        """Joule-Thomson coefficient via central-difference perturbation of enthalpy (K/bar)
        T_arr = np.atleast_1d(np.asarray(T, dtype=float))
        p_arr = np.atleast_1d(np.asarray(p, dtype=float))
        is_scalar = (np.ndim(T) == 0) and (np.ndim(p) == 0)

        EPS = 1e-6
        cp_list = []
        dh_dp_list = []

        for tt, pp in zip(T_arr.flat, p_arr.flat):
            cp_val = float(substance.cp(T=tt, p=pp))
            dp_val = EPS * pp
            h_p = float(substance.h(T=tt, p=pp + dp_val))
            h_m = float(substance.h(T=tt, p=pp - dp_val))
            dh_dp = (h_p - h_m) / (2.0 * dp_val)
            cp_list.append(cp_val)
            dh_dp_list.append(dh_dp)

        mu = -np.array(dh_dp_list) / np.array(cp_list)
        return float(mu[0]) if is_scalar else mu

def n2_temperature_after_expansion(T_in, p_in, p_out):
    """Temperature of N2 after isenthalpic expansion from p_in to p_out."""
    h_in = float(n2.h(T=T_in, p=p_in))
    return float(n2.T(h=h_in, p=p_out))

# Compute temperatures after each expansion stage
T_after_reg = n2_temperature_after_expansion(N2_TEMPERATURE, N2_CYLINDER_PRESSURE, N2_REGULATOR_SET)
JT_drop_reg = N2_TEMPERATURE - T_after_reg

T_after_line = n2_temperature_after_expansion(T_after_reg, N2_REGULATOR_SET, LINE_PRESSURE)
JT_drop_line = T_after_reg - T_after_line

T_final_n2 = T_after_line
JT_drop_total = N2_TEMPERATURE - T_final_n2

# Scan JT coefficient over temperature range 250 K - 350 K at different pressures
T_scan = np.linspace(250, 350, 11)
mu_vs_T_200 = np.array([jt_coeff(n2, T, 200.0) for T in T_scan])
mu_vs_T_3 = np.array([jt_coeff(n2, T, N2_REGULATOR_SET) for T in T_scan])
mu_vs_T_1_5 = np.array([jt_coeff(n2, T, LINE_PRESSURE) for T in T_scan])

print(f"Initial N2 Temperature:      {N2_TEMPERATURE - 273.15:.2f} °C (200.0 bar)")
print(f"After Regulator (3.0 bar):    {T_after_reg - 273.15:.2f} °C (ΔT = -{JT_drop_reg:.2f} °C)")
print(f"In Pipeline (1.5 bar):          {T_final_n2 - 273.15:.2f} °C (Total ΔT = -{JT_drop_total:.2f} °C)")
print(f"μ_JT at 30°C, 200 bar:         {jt_coeff(n2, 303.15, 200.0):.4f} K/bar")
print(f"μ_JT at 30°C, 1.5 bar:         {jt_coeff(n2, 303.15, 1.5):.4f} K/bar")

```

Initial N<sub>2</sub> Temperature: 30.00 °C (200.0 bar)  
 After Regulator (3.0 bar): 0.48 °C (ΔT = -29.52 °C)  
 In Pipeline (1.5 bar): 0.09 °C (Total ΔT = -29.91 °C)  
 μ<sub>JT</sub> at 30°C, 200 bar: 0.0702 K/bar  
 μ<sub>JT</sub> at 30°C, 1.5 bar: 0.2068 K/bar

## 5. Purge Dilution Kinetics & Mixing Models

## Mathematical Formulation of Non-Ideal Mixing

During displacement purging, incoming nitrogen mixes with resident hydrogen. The concentration of residual hydrogen  $C_N$  (% vol) after  $N$  line volume exchanges is modeled as a geometric progression:

$$C_N = C_0(1 - \eta)^N$$

where  $\eta \in (0, 1]$  represents the effective volumetric mixing efficiency factor per volume exchange:

- $\eta = 0.0$  (**Plug Flow**): Pure displacement front with zero mixing. Concentration remains 100% until exactly  $N = 1.0$  volume exchange, at which point  $C \rightarrow 0\%$ .
- $\eta = 1.0$  (**Perfect Mixing**): Equivalent to a Continuous Stirred-Tank Reactor (CSTR). In continuous differential form:

$$\frac{dC}{dN} = -C \implies C(N) = C_0 \cdot e^{-N}$$

- $\eta = 0.3 - 0.7$  (**Real Industrial Purging**): Accounting for buoyancy stratification, velocity profiles, dead legs, and turbulent dispersion.

## Required Volume Exchanges for Safe Ingress

Solving for the required number of line volumes  $N_{\text{req}}$  to reach a target threshold  $C_{\text{target}} = 0.5\%$ :

$$N_{\text{req}} = \frac{\ln\left(\frac{C_{\text{target}}}{C_0}\right)}{\ln(1 - \eta)} \quad (\text{for } 0 < \eta < 1)$$

For ideal mixing ( $\eta = 1.0$ ),  $N_{\text{req}} = \ln\left(\frac{C_0}{C_{\text{target}}}\right)$ .

```
In [5]: def h2_concentration_after_purge(N_volumes, efficiency=0.5):
        """H2 concentration (% vol) after N purge volumes with mixing efficiency η."""
        return INITIAL_H2_PCT * (1.0 - efficiency) ** N_volumes

    def purge_volumes_to_target(efficiency, target=TARGET_H2_PCT):
        """Number of line volumes required to reach target concentration."""
        if efficiency <= 0:
            return float('inf')
        if efficiency >= 1.0:
            return np.log(INITIAL_H2_PCT / target)
        return np.log(target / INITIAL_H2_PCT) / np.log(1.0 - efficiency)

    N_max = 12
    N_volumes = np.arange(0, N_max + 1)

    decay_curves = {eta: h2_concentration_after_purge(N_volumes, eta) for eta in MIXING_EFFICIENCIES}
    purge_vols_needed = {eta: purge_volumes_to_target(eta) for eta in MIXING_EFFICIENCIES}

    print("Purge volumes required to reach ≤ 0.5% H2:")
    for eta in MIXING_EFFICIENCIES:
        print(f"  η = {eta:<3.1f}:  {purge_vols_needed[eta]:.2f} line volumes")
```



Purge volumes required to reach  $\leq 0.5\%$   $H_2$ :

$\eta = 0.3$ : 14.85 line volumes  
 $\eta = 0.5$ : 7.64 line volumes  
 $\eta = 0.7$ : 4.40 line volumes  
 $\eta = 1.0$ : 5.30 line volumes

## 6. Purge Time and Nitrogen Consumption

### Mass Balance & Utility Inventory

The total mass of nitrogen required per line volume exchange depends on real-gas density at pipeline conditions  $\rho_{N_2}(T_{\text{final}}, P_{\text{line}})$ , computed using PYroMat's `mp.N2`:

$$m_{\text{vol}} = \rho_{N_2} \cdot V_{\text{line}}$$

Total nitrogen consumption for  $N_{\text{req}}$  volume exchanges expressed in normal cubic metres ( $Nm^3$ ) at  $P_{\text{norm}} = 1.01325$  bar,  $T_{\text{norm}} = 273.15$  K is:

$$V_{Nm^3} = \frac{N_{\text{req}} \cdot m_{\text{vol}}}{\rho_{\text{norm}}}$$

### Operational Sizing

Standard industrial  $N_2$  cylinders (50 L water capacity at 200 bar) contain approximately  $10 Nm^3$  of usable gas. Evaluating required  $Nm^3$  allows plant engineers to specify cylinder bundle requirements prior to purging.

```
In [6]: # N2 density at line conditions and normal conditions
density_n2_line = float(n2.d(T=T_final_n2, p=LINE_PRESSURE)) # kg/m³
density_n2_norm = float(n2.d(T=T_NORM, p=P_NORM)) # kg/m³

mass_n2_per_volume = density_n2_line * LINE_VOLUME # kg per Line volume

purge_summary_data = []
for eta in MIXING_EFFICIENCIES:
    N_req = purge_vols_needed[eta]
    time_req_min = (N_req * TIME_PER_VOLUME) / 60.0 # minutes
    mass_req_kg = N_req * mass_n2_per_volume # kg
    nm3_req = mass_req_kg / density_n2_norm # Nm³

    purge_summary_data.append({
        'efficiency': eta,
        'volumes': N_req,
        'time_min': time_req_min,
        'mass_kg': mass_req_kg,
        'nm3': nm3_req
    })

print(f"{'η':<6} {'Volumes':<10} {'Time (min)':<12} {'N₂ (kg)':<10} {'N₂ (Nm³)':<10}")
print("-" * 50)
for d in purge_summary_data:
    print(f"{d['efficiency']:<6.1f} {d['volumes']:<10.2f} {d['time_min']:<12.2f} {d['mas
```

| $\eta$ | Volumes | Time (min) | N <sub>2</sub> (kg) | N <sub>2</sub> (Nm <sup>3</sup> ) |
|--------|---------|------------|---------------------|-----------------------------------|
| 0.3    | 14.85   | 21.01      | 24.29               | 19.43                             |
| 0.5    | 7.64    | 10.81      | 12.50               | 10.00                             |
| 0.7    | 4.40    | 6.22       | 7.20                | 5.76                              |
| 1.0    | 5.30    | 7.49       | 8.66                | 6.93                              |

## 7. Visualization & Performance Analysis

### Figure 1: Joule-Thomson Coefficient ( $\mu_{JT}$ ) across Pressures

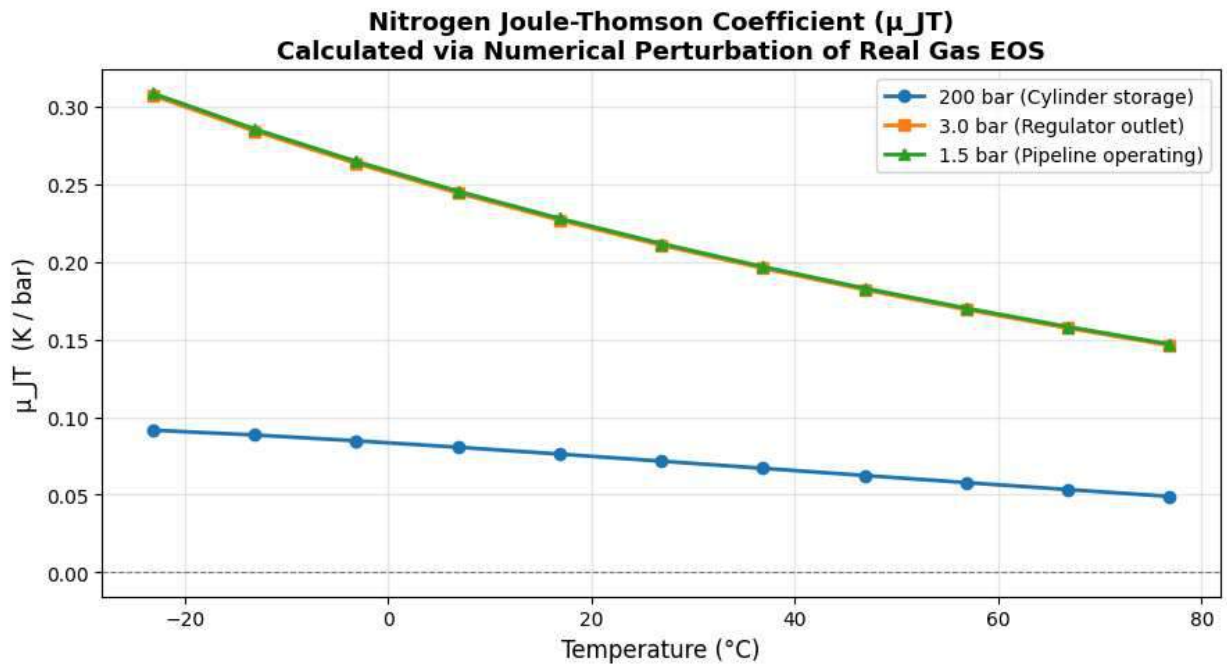
The Joule-Thomson coefficient  $\mu_{JT} = \left( \frac{\partial T}{\partial p} \right)_h$  varies substantially with pressure and temperature:

- At **200 bar** (cylinder manifold), dense-gas repulsive forces reduce  $\mu_{JT}$  to  $\approx 0.07$  K/bar.
- At **3 bar** and **1.5 bar** (line pressure), nitrogen approaches low-density behavior where attractive forces dominate, increasing  $\mu_{JT}$  to  $\approx 0.20 - 0.21$  K/bar.

```
In [7]: SAVE_DIR = '/home/ukhan/engg/ukhan/proj/pyromat_ukhan/'

fig, ax = plt.subplots(figsize=(9, 5))
ax.plot(T_scan - 273.15, mu_vs_T_200, 'o-', color='#1f77b4', linewidth=2,
        label='200 bar (Cylinder storage)')
ax.plot(T_scan - 273.15, mu_vs_T_3, 's-', color='#ff7f0e', linewidth=2,
        label=f'{N2_REGULATOR_SET:.1f} bar (Regulator outlet)')
ax.plot(T_scan - 273.15, mu_vs_T_1_5, '^-', color='#2ca02c', linewidth=2,
        label=f'{LINE_PRESSURE:.1f} bar (Pipeline operating)')
ax.axhline(0, color='gray', linewidth=0.8, linestyle='--')

ax.set_xlabel('Temperature (°C)', fontsize=12)
ax.set_ylabel('μJT (K / bar)', fontsize=12)
ax.set_title('Nitrogen Joule-Thomson Coefficient (μJT)\n'
            'Calculated via Numerical Perturbation of Real Gas EOS',
            fontsize=13, fontweight='bold')
ax.legend(fontsize=10)
ax.grid(True, alpha=0.3)
plt.tight_layout()
#plt.savefig(SAVE_DIR + 'n2_purge_jt_coefficient.jpg', dpi=150)
plt.show()
```



**Figure 2: Temperature Drop Across Expansion Stages**

This figure tracks the thermal history of nitrogen as it expands from 200 bar to line operating pressure (1.5 bar). The primary temperature drop ( $\Delta T \approx -29.5^\circ\text{C}$ ) occurs across the regulator stage (200 bar  $\rightarrow$  3 bar), cooling the gas to near  $0^\circ\text{C}$ .

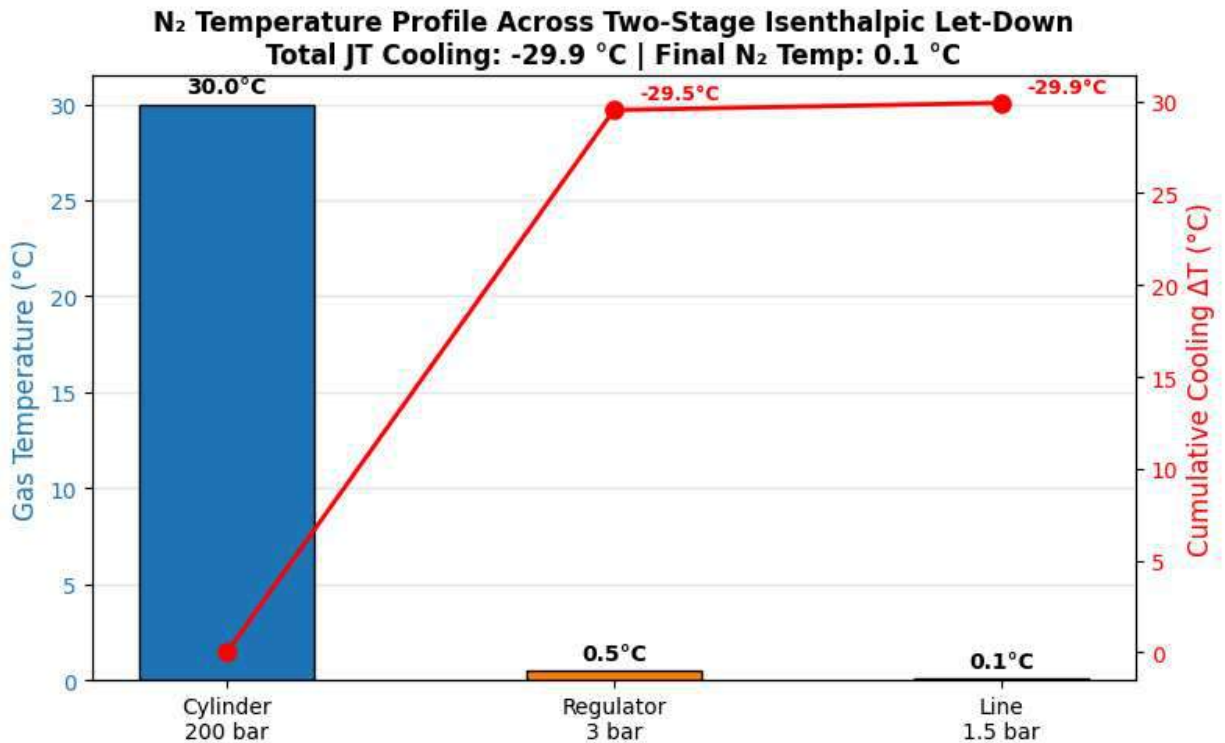
```
In [8]: stages = ['Cylinder\n200 bar', 'Regulator\n3 bar', 'Line\n1.5 bar']
temps = [N2_TEMPERATURE - 273.15, T_after_reg - 273.15, T_final_n2 - 273.15]
drops = [0, JT_drop_reg, JT_drop_total]

fig, ax1 = plt.subplots(figsize=(8, 5))
bars = ax1.bar(stages, temps, color=['#1f77b4', '#ff7f0e', '#2ca02c'],
               edgecolor='black', width=0.45, zorder=3)
ax1.set_ylabel('Gas Temperature (°C)', fontsize=12, color='#1f77b4')
ax1.tick_params(axis='y', labelcolor='#1f77b4')
for bar, val in zip(bars, temps):
    ax1.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.6,
             f'{val:.1f}°C', ha='center', fontsize=10, fontweight='bold')

ax2 = ax1.twinx()
ax2.plot(stages, drops, 'ro-', linewidth=2, markersize=8, zorder=4,
        label='Cumulative ΔT')
ax2.set_ylabel('Cumulative Cooling ΔT (°C)', fontsize=12, color='r')
ax2.tick_params(axis='y', labelcolor='r')
for i, d in enumerate(drops):
    if d != 0:
        ax2.annotate(f'-{d:.1f}°C', (stages[i], d),
                    textcoords='offset points', xytext=(12, 5),
                    fontsize=9, color='r', fontweight='bold')

ax1.set_title('N2 Temperature Profile Across Two-Stage Isenthalpic Let-Down\n'
              f'Total JT Cooling: -{JT_drop_total:.1f} °C | Final N2 Temp: {T_final_n2 - '
              fontsize=12, fontweight='bold')
ax1.grid(True, axis='y', alpha=0.3)
fig.tight_layout()
```

```
#plt.savefig(SAVE_DIR + 'n2_purge_temperature_drop.jpg', dpi=150)
plt.show()
```



**Figure 3: Hydrogen Concentration Decay Kinetics**

Semi-log representation of H<sub>2</sub> concentration vs purge volume exchanges for mixing efficiencies  $\eta \in [0.3, 1.0]$ . Horizontal reference lines mark the LEL (4.0%) and safe hot-work target (0.5%).

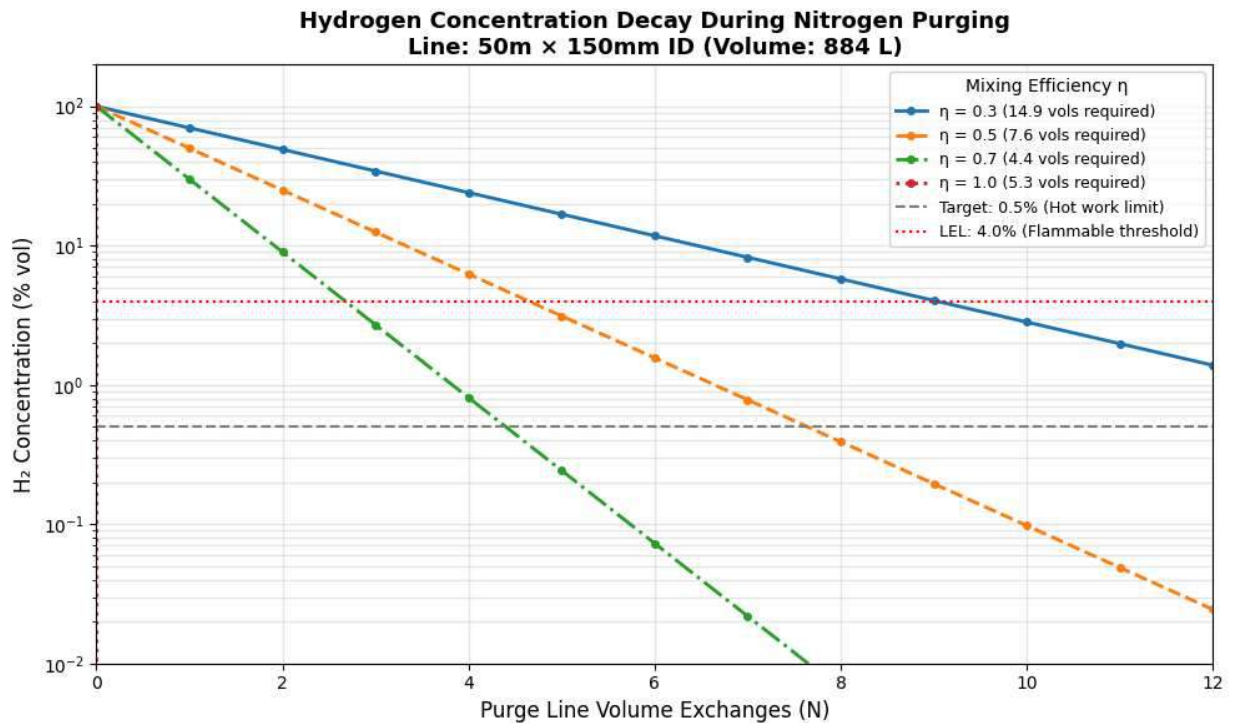
```
In [9]: fig, ax = plt.subplots(figsize=(10, 6))
        colors_eta = ['#1f77b4', '#ff7f0e', '#2ca02c', '#d62728']
        line_styles = ['-', '--', '-.', ':']

        for eta, c, ls in zip(MIXING_EFFICIENCIES, colors_eta, line_styles):
            ax.semilogy(N_volumes, decay_curves[eta], color=c, linestyle=ls,
                        linewidth=2, marker='o', markersize=4,
                        label=f'η = {eta} ({purge_vols_needed[eta]:.1f} vols required)')

        ax.axhline(TARGET_H2_PCT, color='gray', linewidth=1.5, linestyle='--',
                    label=f'Target: {TARGET_H2_PCT}% (Hot work limit)')
        ax.axhline(LEL_H2, color='r', linewidth=1.5, linestyle=':',
                    label=f'LEL: {LEL_H2}% (Flammable threshold)')

        ax.set_xlabel('Purge Line Volume Exchanges (N)', fontsize=12)
        ax.set_ylabel('H2 Concentration (% vol)', fontsize=12)
        ax.set_title('Hydrogen Concentration Decay During Nitrogen Purging\n'
                    f'Line: 50m × 150mm ID (Volume: {LINE_VOLUME_L:.0f} L)',
                    fontsize=13, fontweight='bold')
        ax.legend(fontsize=9, title='Mixing Efficiency η')
        ax.grid(True, alpha=0.3, which='both')
        ax.set_xlim(0, N_max)
        ax.set_ylim(0.01, 200)
        plt.tight_layout()
```

```
#plt.savefig(SAVE_DIR + 'n2_purge_concentration_decay.jpg', dpi=150)
plt.show()
```



**Figure 4: Purge Time and Nitrogen Consumption Sizing**

Comparison of total purging duration (minutes) and nitrogen volume consumed ( $\text{Nm}^3$ ) required to achieve  $\leq 0.5\%$  H<sub>2</sub> across mixing efficiencies.

```
In [10]: fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(13, 5))

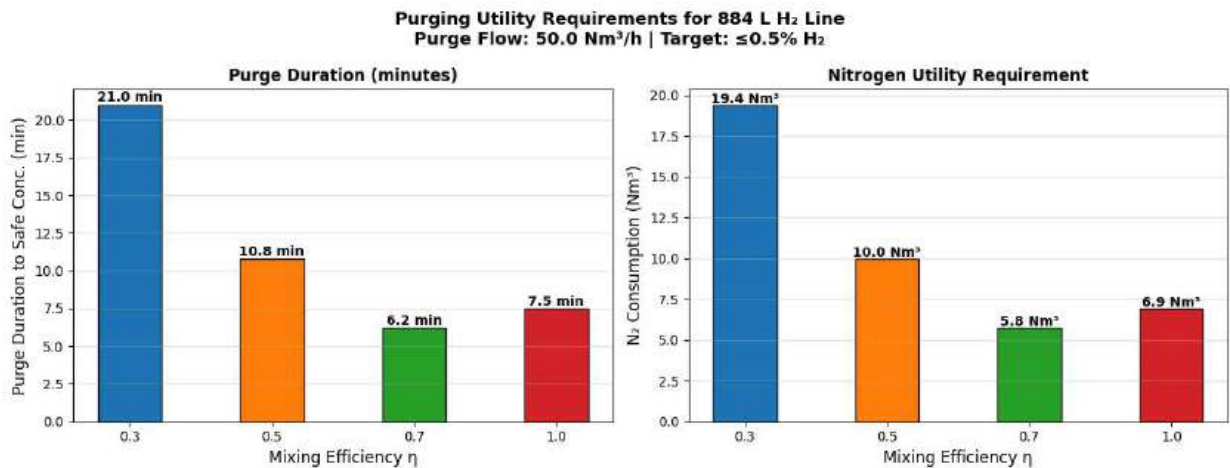
etas_plot = [d['efficiency'] for d in purge_summary_data]
times_plot = [d['time_min'] for d in purge_summary_data]
nm3_plot = [d['nm3'] for d in purge_summary_data]

bars1 = ax1.bar([str(e) for e in etas_plot], times_plot,
                color=colors_eta, edgecolor='black', width=0.45)
ax1.set_xlabel('Mixing Efficiency  $\eta$ ', fontsize=12)
ax1.set_ylabel('Purge Duration to Safe Conc. (min)', fontsize=12)
ax1.set_title('Purge Duration (minutes)', fontsize=12, fontweight='bold')
for bar, val in zip(bars1, times_plot):
    ax1.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.2,
             f'{val:.1f} min', ha='center', fontsize=10, fontweight='bold')
ax1.grid(True, axis='y', alpha=0.3)

bars2 = ax2.bar([str(e) for e in etas_plot], nm3_plot,
                color=colors_eta, edgecolor='black', width=0.45)
ax2.set_xlabel('Mixing Efficiency  $\eta$ ', fontsize=12)
ax2.set_ylabel('N2 Consumption (Nm3)', fontsize=12)
ax2.set_title('Nitrogen Utility Requirement', fontsize=12, fontweight='bold')
for bar, val in zip(bars2, nm3_plot):
    ax2.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.1,
             f'{val:.1f} Nm3', ha='center', fontsize=10, fontweight='bold')
ax2.grid(True, axis='y', alpha=0.3)
```



```
fig.suptitle(f'Purging Utility Requirements for {LINE_VOLUME_L:.0f} L H2 Line\n'
            f'Purge Flow: {PURGE_FLOW_NM3H} Nm3/h | Target: ≤{TARGET_H2_PCT}% H2',
            fontsize=13, fontweight='bold')
plt.tight_layout()
#plt.savefig(SAVE_DIR + 'n2_purge_time_consumption.jpg', dpi=150)
plt.show()
```



**Figure 5: Purge Duration Sensitivity Contour Map**

Parametric sensitivity map of purge time as a function of pipeline volume (10 – 1000 L) and mixing efficiency ( $\eta = 0.2 - 1.0$ ) at 50 Nm<sup>3</sup>/h purge rate.

```
In [11]: line_volumes = np.logspace(1, 3, 50) # 10 to 1000 Litres
efficiencies = np.linspace(0.2, 1.0, 50)
V_grid, E_grid = np.meshgrid(line_volumes, efficiencies)

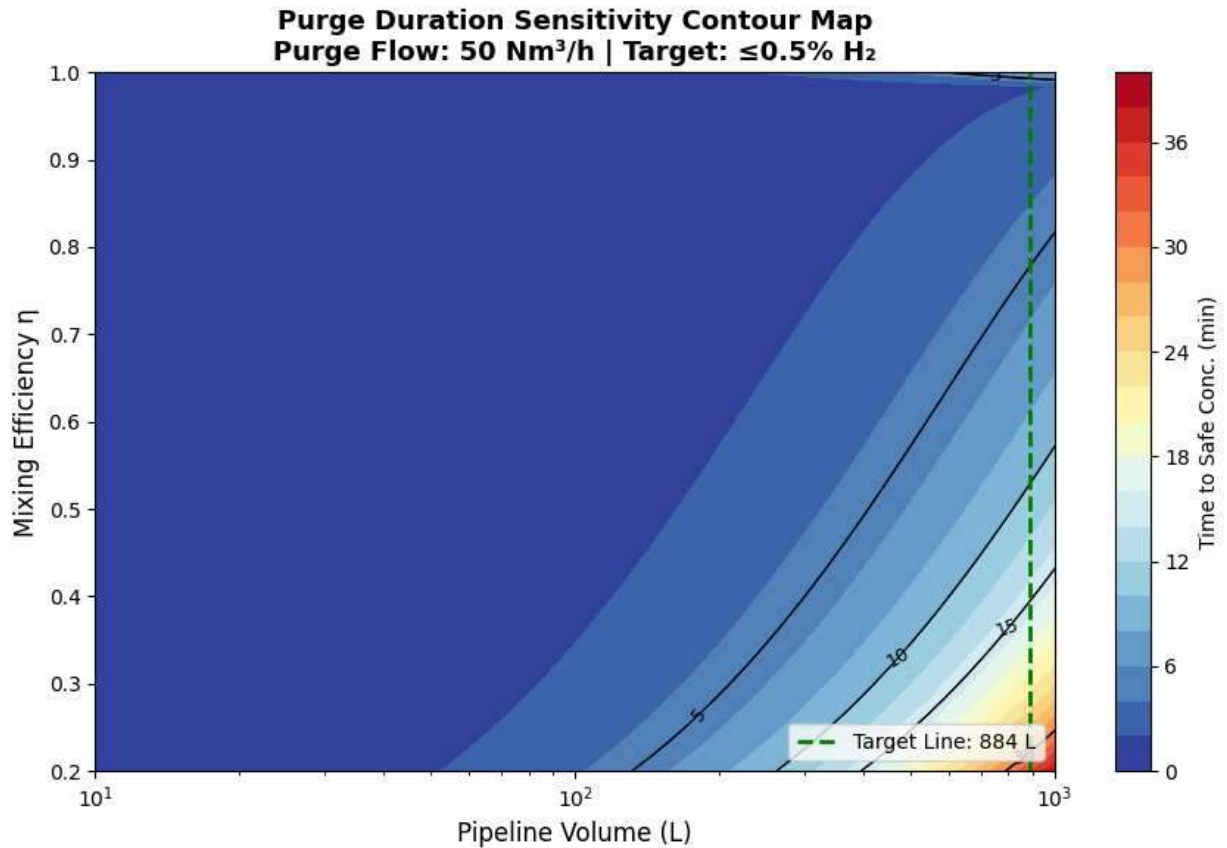
with np.errstate(divide='ignore'):
    N_req_grid = np.where(E_grid < 1.0,
                          np.log(TARGET_H2_PCT / INITIAL_H2_PCT) / np.log(1.0 - E_grid),
                          np.log(INITIAL_H2_PCT / TARGET_H2_PCT))

    flow_actual_grid = (PURGE_FLOW_NM3H / 3600.0) * (P_NORM / LINE_PRESSURE) * (LINE_TEMP /
    time_grid = N_req_grid * (V_grid / 1000.0) / flow_actual_grid / 60.0

fig, ax = plt.subplots(figsize=(9, 6))
contour = ax.contourf(V_grid, E_grid, time_grid, levels=20, cmap='RdYlBu_r')
cbar = plt.colorbar(contour, ax=ax, label='Time to Safe Conc. (min)')
ax.contour(V_grid, E_grid, time_grid, levels=[5, 10, 15, 30, 60],
           colors='k', linewidths=0.8)
ax.clabel(ax.contour(V_grid, E_grid, time_grid, levels=[5, 10, 15, 30, 60],
                  colors='k', linewidths=0.8), inline=True, fontsize=9)
ax.axvline(LINE_VOLUME_L, color='g', linewidth=2, linestyle='--',
           label=f'Target Line: {LINE_VOLUME_L:.0f} L')

ax.set_xlabel('Pipeline Volume (L)', fontsize=12)
ax.set_ylabel('Mixing Efficiency  $\eta$ ', fontsize=12)
ax.set_title('Purge Duration Sensitivity Contour Map\n'
            'Purge Flow: 50 Nm3/h | Target: ≤0.5% H2',
            fontsize=13, fontweight='bold')
ax.legend(loc='lower right')
ax.set_xscale('log')
plt.tight_layout()
```

```
#plt.savefig(SAVE_DIR + 'n2_purge_sensitivity.jpg', dpi=150)
plt.show()
```



**Figure 6: Operational Safety Envelope & Hazard Classification**

Tri-zone safety classification envelope mapping purge progress:

- **Red Zone** ( $H_2 > 4.0\%$ ): Flammable / explosive regime. Hot work prohibited.
- **Yellow Zone** ( $0.5\% < H_2 \leq 4.0\%$ ): Marginal region. Continuous monitoring required.
- **Green Zone** ( $H_2 \leq 0.5\%$ ): Safe operating threshold for welding / hot work.

```
In [12]: N_detailed = np.linspace(0, N_max, 250)
fig, ax = plt.subplots(figsize=(10, 6))

ax.axhspan(0, SAFE_MARGIN * LEL_H2, alpha=0.15, color='green',
           label=f'Safe Zone ( $H_2 \leq \{SAFE\_MARGIN*LEL\_H2:.2f\}\%$ )')
ax.axhspan(SAFE_MARGIN * LEL_H2, LEL_H2, alpha=0.15, color='gold',
           label=f'Marginal Warning ( $\{SAFE\_MARGIN*LEL\_H2:.2f\}\% - \{LEL\_H2:.1f\}\%$ )')
ax.axhspan(LEL_H2, 20.0, alpha=0.15, color='red',
           label=f'EXPLOSIVE HAZARD ( $H_2 > \{LEL\_H2:.1f\}\%$ )')

for eta, c, ls in zip(MIXING_EFFICIENCIES, colors_eta, line_styles):
    conc = h2_concentration_after_purge(N_detailed, eta)
    ax.plot(N_detailed, conc, color=c, linestyle=ls, linewidth=2, label=f' $\eta = \{eta\}$ ')

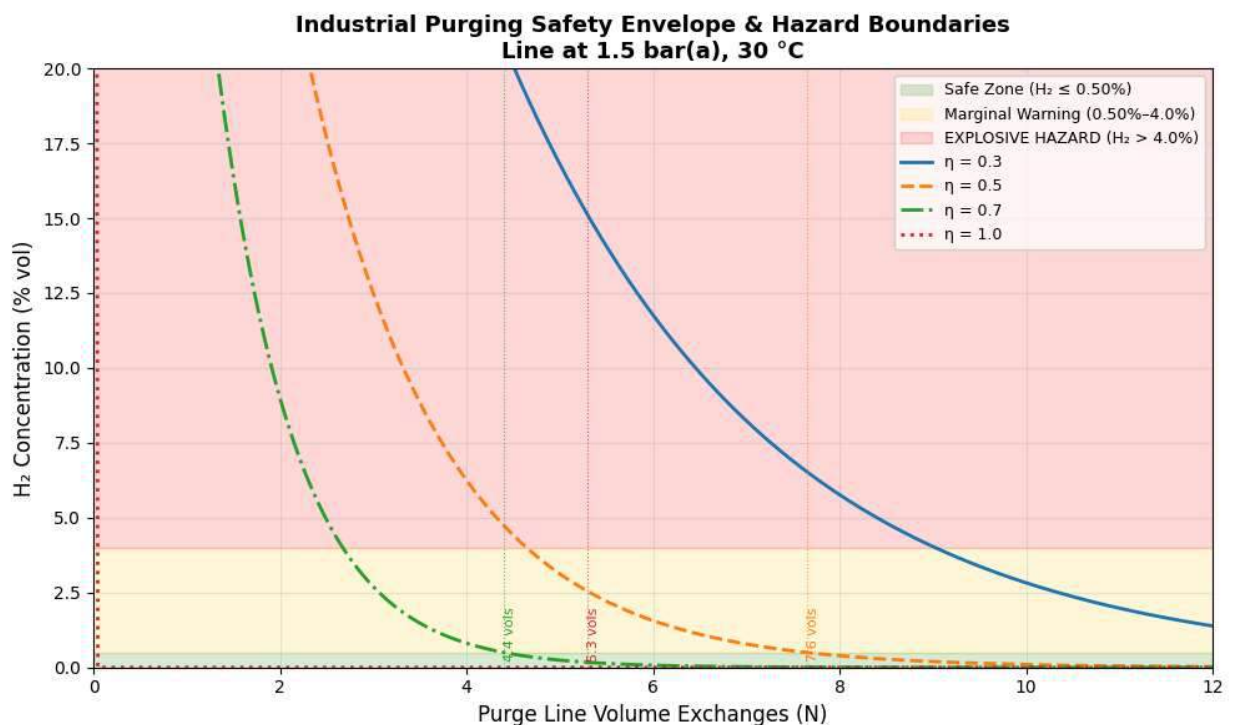
for eta in MIXING_EFFICIENCIES:
    N_safe = purge_volumes_to_target(eta, SAFE_MARGIN * LEL_H2)
    if np.isfinite(N_safe) and N_safe <= N_max:
        ax.axvline(N_safe, color=colors_eta[MIXING_EFFICIENCIES.index(eta)],
```

```

        linewidth=0.8, linestyle=':', alpha=0.7)
    ax.annotate(f'{N_safe:.1f} vols', (N_safe, 0.2),
               fontsize=8, color=colors_eta[MIXING_EFFICIENCIES.index(eta)],
               rotation=90, va='bottom')

    ax.set_xlabel('Purge Line Volume Exchanges (N)', fontsize=12)
    ax.set_ylabel('H2 Concentration (% vol)', fontsize=12)
    ax.set_title('Industrial Purging Safety Envelope & Hazard Boundaries\n'
                 'Line at 1.5 bar(a), 30 °C',
                 fontsize=13, fontweight='bold')
    ax.legend(fontsize=9, loc='upper right')
    ax.set_ylim(0, 20)
    ax.set_xlim(0, N_max)
    ax.grid(True, alpha=0.3)
    plt.tight_layout()
    #plt.savefig(SAVE_DIR + 'n2_purge_safety_envelope.jpg', dpi=150)
    plt.show()

```



## 8. Engineering Recommendations & Operational Protocol

### Recommended Industrial Purging Protocol

| Phase                 | Action           | Process Detail                                                                          |
|-----------------------|------------------|-----------------------------------------------------------------------------------------|
| 1. Isolation          | Line Lockout     | Close and tag double-block-and-bleed valves at both line battery limits.                |
| 2. Depressurization   | Atmospheric Vent | Open elevated vent stack to reduce pressure to atmospheric ( 1.0 bar(a)).               |
| 3. Utility Connection | Regulated Supply | Connect N <sub>2</sub> manifold at injection point with pre-heater or staged regulator. |

| Phase              | Action              | Process Detail                                                                                                                |
|--------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------|
| 4. Purge Execution | Flow Control        | Maintain velocity $> 5$ m/s to ensure turbulent mixing ( $\eta \approx 0.7$ ). Purge minimum <b>4.5 to 7.0 line volumes</b> . |
| 5. Verification    | Atmospheric Testing | Sample vent gas with calibrated electrochemical / thermal conductivity $H_2$ meter until $C_{H_2} \leq 0.5\%$ .               |
| 6. Maintenance     | Positive Pressure   | Maintain 0.1 – 0.2 bar positive $N_2$ purge pad during hot work to prevent air ingress.                                       |

## Material Integrity & JT Cooling Mitigation

- **MDMT Concern:** Uncontrolled let-down yields  $N_2$  at  $-0.1^\circ\text{C}$  to  $-29.5^\circ\text{C}$ . Verify line material spec against Minimum Design Metal Temperature (MDMT for standard A106 Gr B carbon steel without impact testing is  $-29^\circ\text{C}$ ).
- **Mitigation:** Use an inline electric or steam heater upstream of the regulator during high-flow purging.

```
In [13]: print("=" * 65)
print("  N2 PURGE SIMULATION — HYDROGEN LINE INERTING SUMMARY")
print("=" * 65)

print(f"\nLINE SPECIFICATIONS")
print(f"  Length:                {LINE_LENGTH:.0f} m")
print(f"  Diameter:              {LINE_DIAMETER*1000:.0f} mm  ({LINE_DIAMETER*39.37:.1f} i")
print(f"  Volume:                {LINE_VOLUME_L:.0f} L  ({LINE_VOLUME:.3f} m³)")
print(f"  Operating Pressure:    {LINE_PRESSURE:.1f} bar(a)")
print(f"  Temperature:          {LINE_TEMP-273.15:.0f} °C")

print(f"\nJOULE-THOMSON COOLING (200 → 1.5 bar)")
print(f"  After Regulator:       {T_after_reg-273.15:.1f} °C  (ΔT = -{JT_drop_reg:.1f} °C)")
print(f"  In Pipeline:          {T_final_n2-273.15:.1f} °C  (Total ΔT = -{JT_drop_total:.1f} °C)")
print(f"  μJT at 30°C:          {jt_coeff(n2, 303.15, 200.0):.4f} K/bar (200 bar)")
print(f"                        {jt_coeff(n2, 303.15, 1.5):.4f} K/bar (1.5 bar)")

print(f"\nPURGE FLOW & UTILITIES")
print(f"  Normal Flow Rate:      {PURGE_FLOW_NM3H} Nm³/h")
print(f"  Actual Flow Rate:      {PURGE_FLOW_ACTUAL:.4f} m³/s  ({PURGE_FLOW_ACTUAL*1000:.1f} L/s)")
print(f"  Time per Volume:       {TIME_PER_VOLUME:.1f} s")
print(f"  N2 Density in Line:   {density_n2_line:.3f} kg/m³")

print(f"\nDILUTION SUMMARY TO ≤ {TARGET_H2_PCT}% H2")
print(f"  {'η':<6} {'Volumes':<10} {'Time(min)':<10} {'N2(kg)':<10} {'N2(Nm³)':<10} Stat")
print(f"  {'-'*55}")
for d in purge_summary_data:
    status = '✓ Feasible' if d['volumes'] < 20 else 'X Inefficient'
    print(f"    {d['efficiency']:<6.1f} {d['volumes']:<10.1f} {d['time_min']:<10.1f} "
          f"{d['mass_kg']:<10.2f} {d['nm3']:<10.1f} {status}")

best_purge = purge_summary_data[2]  # η = 0.7
print(f"\nRECOMMENDED OPERATING POINT (η = 0.7)")
print(f"  • Recommended Purge:   {best_purge['volumes']:.1f} line volumes")
print(f"  • Expected Duration:   {best_purge['time_min']:.1f} minutes")
print(f"  • N2 Requirement:     {best_purge['nm3']:.1f} Nm³ (~1 cylinder bundle)")
```

```
print(" • Verification:      Portable electrochemical H2 detector at vent")
print(f" • Temperature Note:  Gas enters line at {T_final_n2-273.15:.1f}°C. Verify MDM")
```

```
=====
N2 PURGE SIMULATION – HYDROGEN LINE INERTING SUMMARY
=====
```

LINE SPECIFICATIONS

Length: 50 m  
Diameter: 150 mm (5.9 in)  
Volume: 884 L (0.884 m<sup>3</sup>)  
Operating Pressure: 1.5 bar(a)  
Temperature: 30 °C

JOULE-THOMSON COOLING (200 → 1.5 bar)

After Regulator: 0.5 °C ( $\Delta T = -29.5$  °C)  
In Pipeline: 0.1 °C (Total  $\Delta T = -29.9$  °C)  
 $\mu_{JT}$  at 30°C: 0.0702 K/bar (200 bar)  
0.2068 K/bar (1.5 bar)

PURGE FLOW & UTILITIES

Normal Flow Rate: 50.0 Nm<sup>3</sup>/h  
Actual Flow Rate: 0.0104 m<sup>3</sup>/s (10.4 L/s)  
Time per Volume: 84.9 s  
N<sub>2</sub> Density in Line: 1.851 kg/m<sup>3</sup>

DILUTION SUMMARY TO  $\leq 0.5\%$  H<sub>2</sub>

| $\eta$ | Volumes | Time(min) | N <sub>2</sub> (kg) | N <sub>2</sub> (Nm <sup>3</sup> ) | Status     |
|--------|---------|-----------|---------------------|-----------------------------------|------------|
| 0.3    | 14.9    | 21.0      | 24.29               | 19.4                              | ✓ Feasible |
| 0.5    | 7.6     | 10.8      | 12.50               | 10.0                              | ✓ Feasible |
| 0.7    | 4.4     | 6.2       | 7.20                | 5.8                               | ✓ Feasible |
| 1.0    | 5.3     | 7.5       | 8.66                | 6.9                               | ✓ Feasible |

RECOMMENDED OPERATING POINT ( $\eta = 0.7$ )

- Recommended Purge: 4.4 line volumes
- Expected Duration: 6.2 minutes
- N<sub>2</sub> Requirement: 5.8 Nm<sup>3</sup> (~1 cylinder bundle)
- Verification: Portable electrochemical H<sub>2</sub> detector at vent
- Temperature Note: Gas enters line at 0.1°C. Verify MDMT of piping.